

Creating and Analyzing Surveys



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INSH 6300
Fall 2026

Agenda

- Introduction to Qualtrics Forms for Survey Creation
- Introduction to LimeSurvey
- Digitization of paper survey responses
- Exporting and Analyzing Surveys from Qualtrics
- Computation text analysis of open responses
- Slides, handouts, and sample survey available at:

Sample: Campus Commute

Take a few minutes to fill out this survey. We will use this to point to examples: <https://bit.ly/fa26-scott-survey>

What did you notice?

Research Ethics

While doing official research involving human subjects, always:

1. Get IRB ([Institutional Review Board](#)) approval
2. Find the right, secure survey tool to use; Northeastern uses Qualtrics, which is secure for keeping your results private
3. Understand that your specific questions will gather specific answers. Try not to steer your participants in a particular direction, and always be wary of the vulnerabilities of the demographic(s) you're studying.

Why choose Qualtrics?

Pros

- Access through institutional affiliation
- Ability to create up to 100-question surveys
- Quick insights
- Built-in sentiment analysis
- Multi-channel distribution (web, email, SMS, app, chatbot)

Cons

- High cost of subscription
- No ability for self-hosting
- Steep learning curve

Qualtrics

- Qualtrics is a great survey option for more formal surveys.
- You can use it in many industries: political campaigns, marketing, academia and many others.
- It has many pre-programmed, built-in questions which can save you a lot of time.

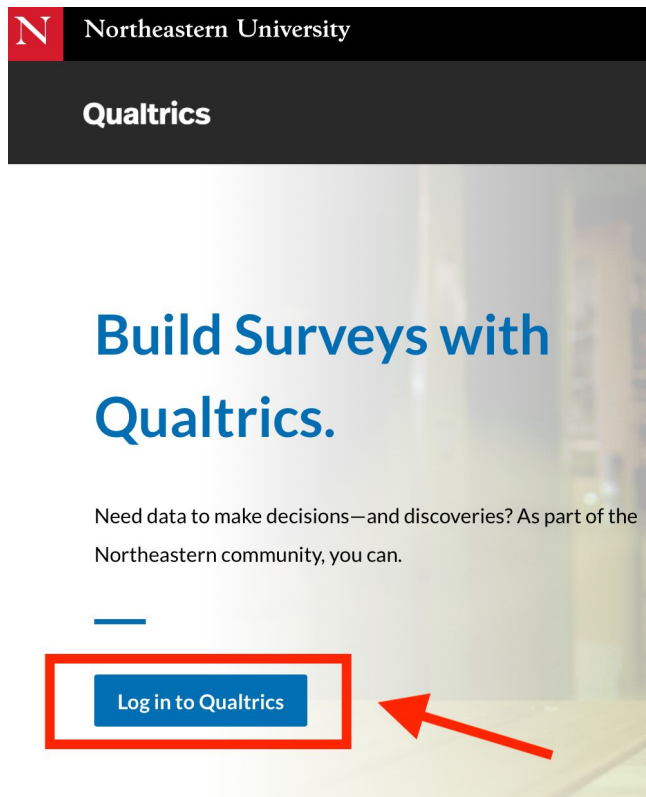


- You can login with your Northeastern credentials: qualtrics.northeastern.edu

Once in Qualtrics

Step 1. Log in with your Northeastern credentials.

<https://qualtrics.northeastern.edu/>



Step 2. Begin the project

The screenshot shows the XM dashboard interface. At the top left, there is a navigation bar with the XM logo, a menu icon, and the word 'Home'. Below this, a 'Welcome to XM' section features a user profile icon with the letter 'Y' and a settings gear icon. A search bar is present with the placeholder text 'Search by name, type, owner...'. The main content area is divided into two columns. The left column, titled 'Recently visited', lists three survey items: 'Survey: Commuting Habits of Northe...' (with a 'New' button), 'Survey: Flex pilot survey for FACEB...' (with '23 Responses' and an 'Active' status), and 'Survey: My Sample Survey' (with a 'New' button). Below these is another survey item: 'Survey: Facebook Warehouse Biling...' (with '1354 Responses' and an 'Active' status). At the bottom of this column is a blue button labeled 'Create a new project'. The right column, titled 'Active surveys', displays three survey cards. The first card is 'Warehouse pilot survey' (0 new responses, 104 total responses). The second card is 'Introduction & Team Formation Form - Fa...' (0 new responses, 43 total responses). The third card is 'Facebook Warehouse E' (0 new responses, 1.4K total responses). Each card has a star icon in the top right corner. Below the 'Active surveys' section is a 'Your workflows summary' section with a dashed border, containing a document icon, the text 'Set up your first workflow to automate important actions and drive efficiency', and two buttons: 'Create a workflow' and 'Learn more about workflows'. Red arrows point from text boxes to specific elements: one from the 'Warehouse pilot survey' card to the 'Active surveys' title, another from the 'See all projects' link to the 'Recently visited' title, and a third from the 'Create a new project' button to the 'Your workflows summary' section.

Welcome to XM

Search by name, type, owner...

Recently visited [See all projects](#)

Survey
Commuting Habits of Northe... [New](#)

Survey
Flex pilot survey for FACEB...
23 Responses [Active](#)

Survey
My Sample Survey [New](#)

Survey
Facebook Warehouse Biling...
1354 Responses [Active](#)

[Create a new project](#)

Active surveys

Warehouse pilot survey
0 new responses
104 total responses

Introduction & Team Formation Form - Fa...
0 new responses
43 total responses

Facebook Warehouse E
0 new re:
1.4K total responses

Your workflows summary

Set up your first workflow to automate important actions and drive efficiency

[Create a workflow](#) [Learn more about workflows](#)

You will see your active and recently visited surveys.

Start by clicking "Create a new project"

Step 3. Open a blank project



Search the catalog

Type

What you're measuring

Customers

Employees

Markets

Students

Website and app visitors

Department

Customer experience

Education

Human resources

Create a project

From scratch



Survey

Guided projects

Start building using a pre-built solution with step-by-step guidance



Concept Testing (Basic)

Assess the strengths and weaknesses of potential new products or services



Pricing Study (Van Westendorp)

Identify optimal product pricing with 4 simple questions



Product Naming

Find the best name product

Once you begin, you will have the option to either start from scratch or import the pre-built survey options.

Step 4. You can:

- Name your survey right away:
- Place it in a folder (especially, if you have several survey projects)
- Start a blank survey or export the pre-built survey questions. A blank survey option is the most common.

Create a new project

Survey

Name

My Sample Survey

Folder

All Projects & Programs

How do you want to start your survey?

Create a blank survey project

Create project

Cancel

Finally, begin the project by clicking here.

Basic Anatomy of Qualtrics

The screenshot displays the Qualtrics web application interface. At the top, there is a navigation bar with the Qualtrics logo (XM) and a menu icon. Below this, a horizontal tab bar contains several tabs: 'Survey', 'Workflows', 'Distributions', 'Data & Analysis', 'Results', and 'Reports'. The 'Survey' tab is currently selected and highlighted with a red underline. A red oval encircles the 'Survey' tab and the vertical sidebar on the left. The sidebar contains four icons: a clipboard (selected), a document, a question mark, and a person. A red arrow points from the 'Survey' tab to the sidebar. Another red arrow points from the 'Reports' tab to a text box. The main content area shows the 'Edit question' screen for an 'Untitled project'. It includes a 'Question type' dropdown set to 'Multiple choice', an 'Answer type' section with radio buttons for 'Allow one answer' (selected) and 'Allow multiple answers', and a 'Choices' section with a 'Number of choices' set to 3. Below these are buttons for 'Import from library' and 'Add new question'. At the bottom, there is a section for 'End of Survey' with a message: 'We thank you for your time spent taking this survey. Your response has been recorded.'

Once the survey is ready, you will distribute it and analyze the subsequent results in these section (all horizontal tabs)

All major survey design options (question type, the # of answer choices, etc.) will be located here, in the "Survey" tab and all of the vertical tabs.

Blocks and Questions

You will see spaces for a survey title, block title, questions and answer choices.

dit question

uestion type

Multiple choice

Answer type

Allow one answer

Allow multiple answers

Choices

umber of choices

3

dit multiple

se suggested choices

Tools

Saved at 6:33 PM

Draft

Communting Habits of Northeastern Students

Block Title

Question

Text of the question

Click to write Choice 1

Click to write Choice 2

Click to write Choice 3

Import from library

Add Block

Survey and block titles are visible to you only!

Different Types of Questions

Multiple choice: questions that only have specific answers and the user can only click one. For example, “**What is your favorite book genre?**”

Likert-scale: a type of response scale in which respondents can specify their level of agreement, importance or satisfaction typically in 5 points, (1) strongly agree to (5) strongly disagree.

Checkboxes: questions that allow user to select multiple options from a set of responses. For example, “select all forms of transportation that you take to campus”.

Linear Scale: questions that invite users to choose from a lower to higher number that match their experience. Typically, on a scale from 1 to 10; allows a more granular measure of affect because participants are able to express their degrees of response.

Paragraph/Short Answer: open-ended questions that usually invite longer prose. For example, “Describe your favorite gym exercise and how it makes you feel”.

Evaluate the following questions (1/3)

- *On a scale from 1 to 10 please rate the discounted semester pass program.*
 - **Problem:** Jargon/assumed knowledge. Respondents may not know the program or be unfamiliar with this term. Don't assume familiarity.
- *Do you think that many programs that are currently reviewed by the university administration, like the expansion of bike lanes on campus, would be environmentally friendly and affordable?*
 - **Problem:** Vaguely worded and asks about two things at once but only allows for one answer. Make sure to ask about one thing at a time using plain language.

Evaluate the following questions (2/3)

- *Do you think the inept university administration should for once finally do something about improving our campus commuting experience?*
 - **Problem:** Biased language. Make sure to keep wording neutral and avoid terms like “inept” or “for once” that signal a desired answer.
- *How long does it take you to commute to campus on a typical day? Less than 10 minutes / 11–20 minutes / 15–35 minutes / 25–40 minutes / 41–50 minutes / 51–60 minutes / More than one hour*
 - **Problem:** Overlapping and inconsistent answer ranges. When options overlap, responses will vary. Make sure to use mutually exclusive ranges so that there are no gaps or overlap.

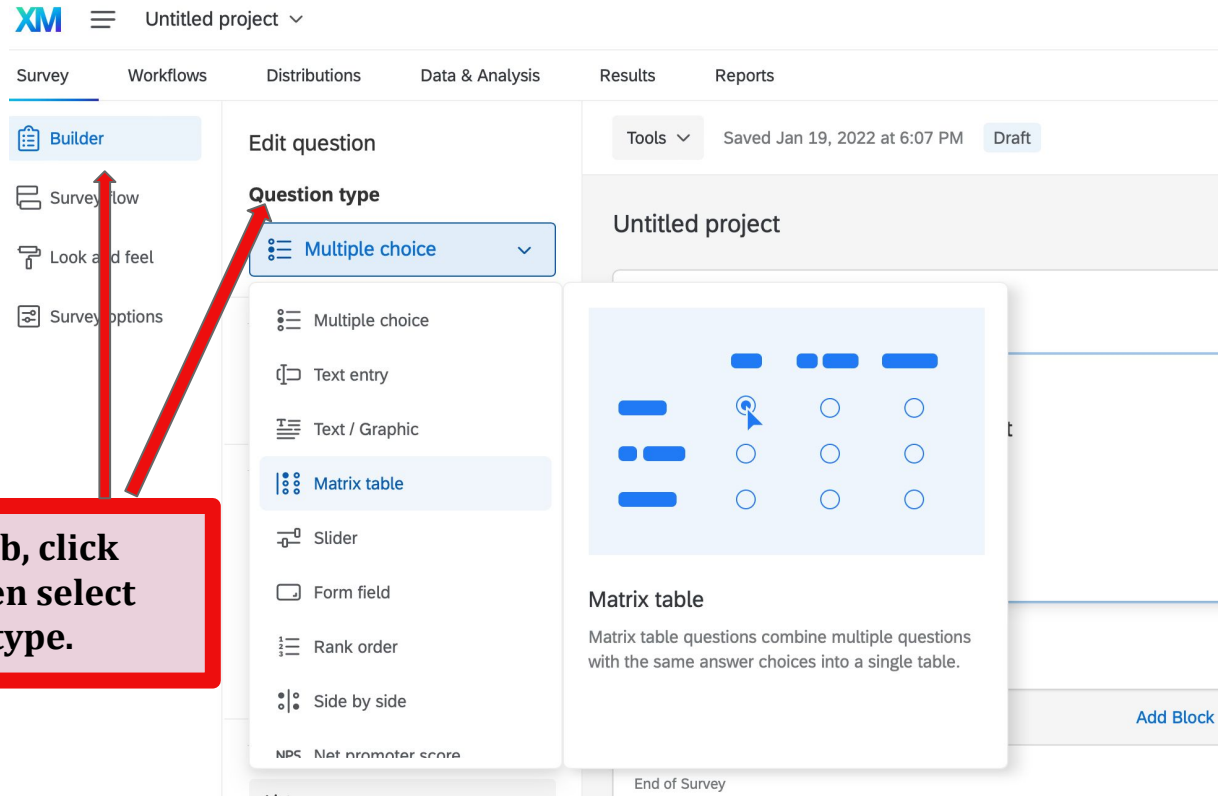
Evaluate the following questions (3/3)

- *Do you always take the MBTA when you commute to campus? Choices: Sometimes, Rarely, Never*
 - **Problem:** The questions and answers do not match. The question includes "always," but it is not represented in the choices. Make sure that the responses match what is being asked.
- *Does not the MBTA provide adequate resources for commuters? Choices: Yes/No*
 - **Problem:** Double negative. Phrasing like “does not...adequate” is confusing to survey takers since it is not clear what “yes” means.

Answer Choices in Qualtrics

- You can find the common question types we just discussed and many many more in Survey Builder. Here is the example of a Likert scale question.

In “Survey” tab, click “Builder”. Then select the question type.



Answer Choices in Qualtrics. Likert Scale

Choices

Number of choices

— 5 +

[Edit multiple](#)

Use suggested choices ☒

Dissatisfied - Satisfied

☒ Reverse order

Choice display

☒ Show all

☐ Show first and last

Format

List

[Alignment](#)

Horizontal

[Add Block](#)

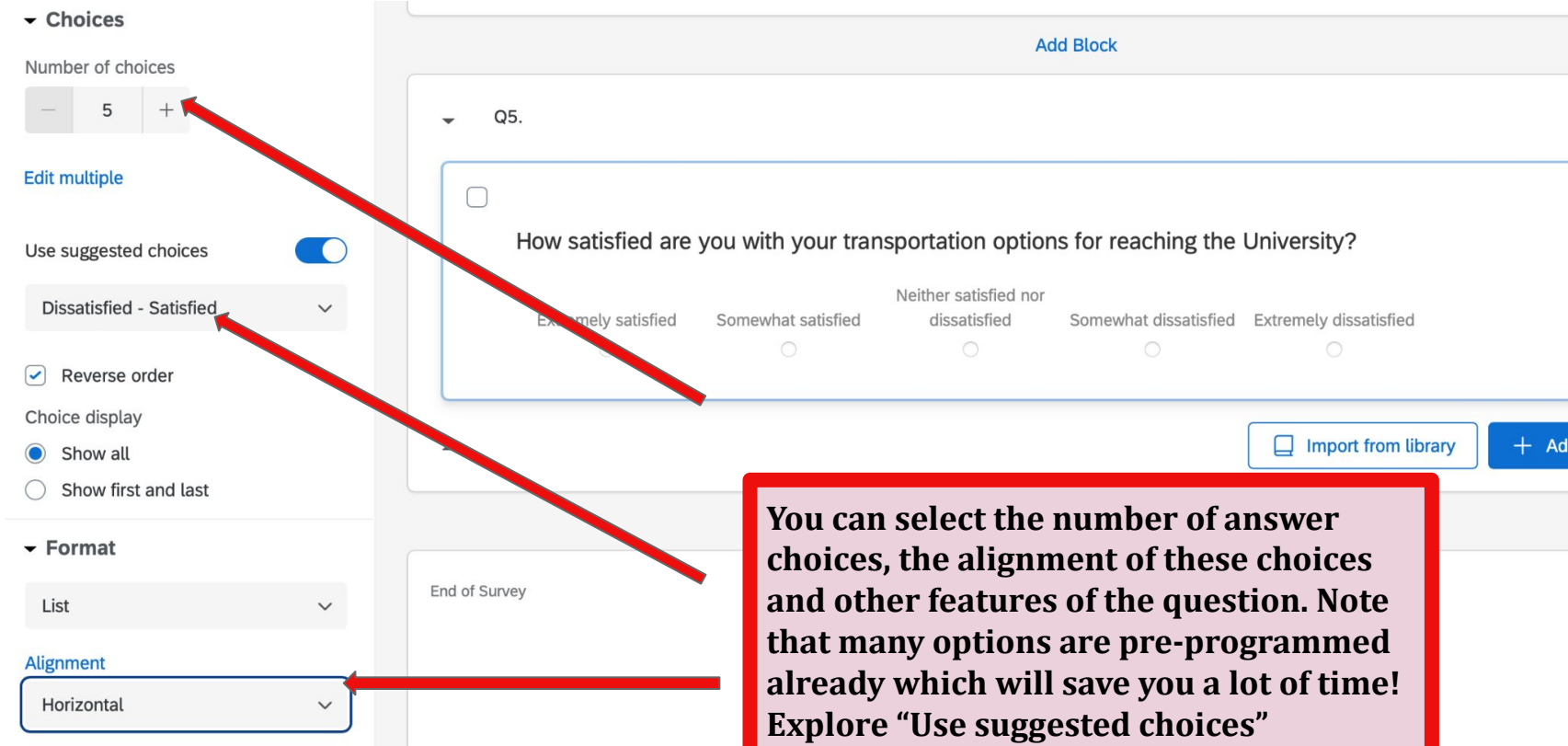
Q5.

☐ How satisfied are you with your transportation options for reaching the University?

Extremely satisfied Somewhat satisfied Neither satisfied nor dissatisfied Somewhat dissatisfied Extremely dissatisfied

[Import from library](#) [+ Add](#)

End of Survey



You can select the number of answer choices, the alignment of these choices and other features of the question. Note that many options are pre-programmed already which will save you a lot of time! Explore “Use suggested choices”

Multiple choice

Tools Saved at 5:25 PM Draft

Answer type

☒ Allow one answer

☐ Allow multiple answers

Choices

Number of choices

5

Edit multiple

Use suggested choices ☒

Dissatisfied - Satisfied

Disagree - Agree

Dissatisfied - Satisfied

Inappropriate - Appropriate

Male - Female - Non-binary

True - False

Yes - No

11-20 minutes

21-30 minutes

31-40 minutes

41-50 minutes

51-60 minutes

More than one hour

☒ transit_optns_satis

How satisfied are you with your transportation options for reaching the

☐ Extremely dissatisfied

☐ Somewhat dissatisfied

☐ Neither satisfied nor dissatisfied

☐ Somewhat satisfied

☐ Extremely satisfied

Programming in Qualtrics

By labelling your questions (shorter titles preferred), you will save a lot of time in the analysis stage. Instead of question 1, question 2, etc. you will have meaningful labels that you can then add to your codebook.

Note how by selecting "Use suggested choices" you can automatically populate your answer choices

Collaborating in Qualtrics

The screenshot shows the Qualtrics web interface. On the left, the 'Edit question' sidebar is visible with 'Question type' set to 'Multiple choice' and 'Answer type' set to 'Allow one answer'. The main area shows a question block with a 'Tools' dropdown menu open. The 'Tools' menu includes options like 'Auto-number questions', 'Reset recode values', 'Manage reusable choices', 'Generate test responses', 'Review', 'Collaborate', 'Versions', and 'Import/Export'. The 'Collaborate' option is highlighted. A red box with the text 'Step 1. Click on the “Tools” icon. Then select “Collaborate”.' and a red arrow points to the 'Tools' icon and the 'Collaborate' option. The top navigation bar includes 'Survey', 'Workflows', 'Distributions', 'Data & Analysis', 'Results', and 'Reports'. The top right corner has a search icon, 'Preview', 'Publish' buttons, and a user profile icon.

Collaborating in Qualtrics

Collaborate on Project [Redacted]

Type Username or Email... Add [User and Group Address Book](#)

User	Edit	View Reports	Activate/Deactivate	Copy	Distribute
Hannah Johnston [Redacted]	☒ Details	☒ Details	☒	☒	☒

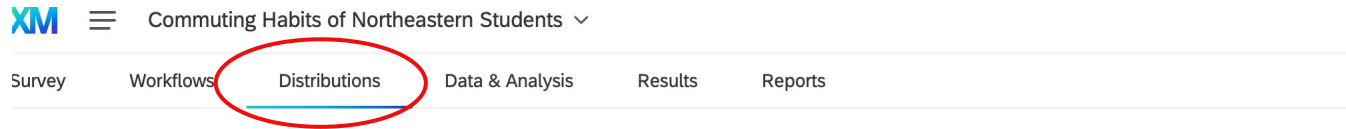
Cancel Save

Step 2. Type the names of your project collaborators and click "Add".

If you want collaborators to have full access to the survey design (editing, copying, etc), make sure you check these boxes.

Survey Distribution

- Once the survey is ready, there are multiple ways in which you can distribute it, including web-links and QR codes.



Go to Distributions and select the method you prefer.

How do you want to distribute your survey?



Email



Web



Social



Mobile



Online panel

Send with Qualtrics



Compose Email

Use your own email system



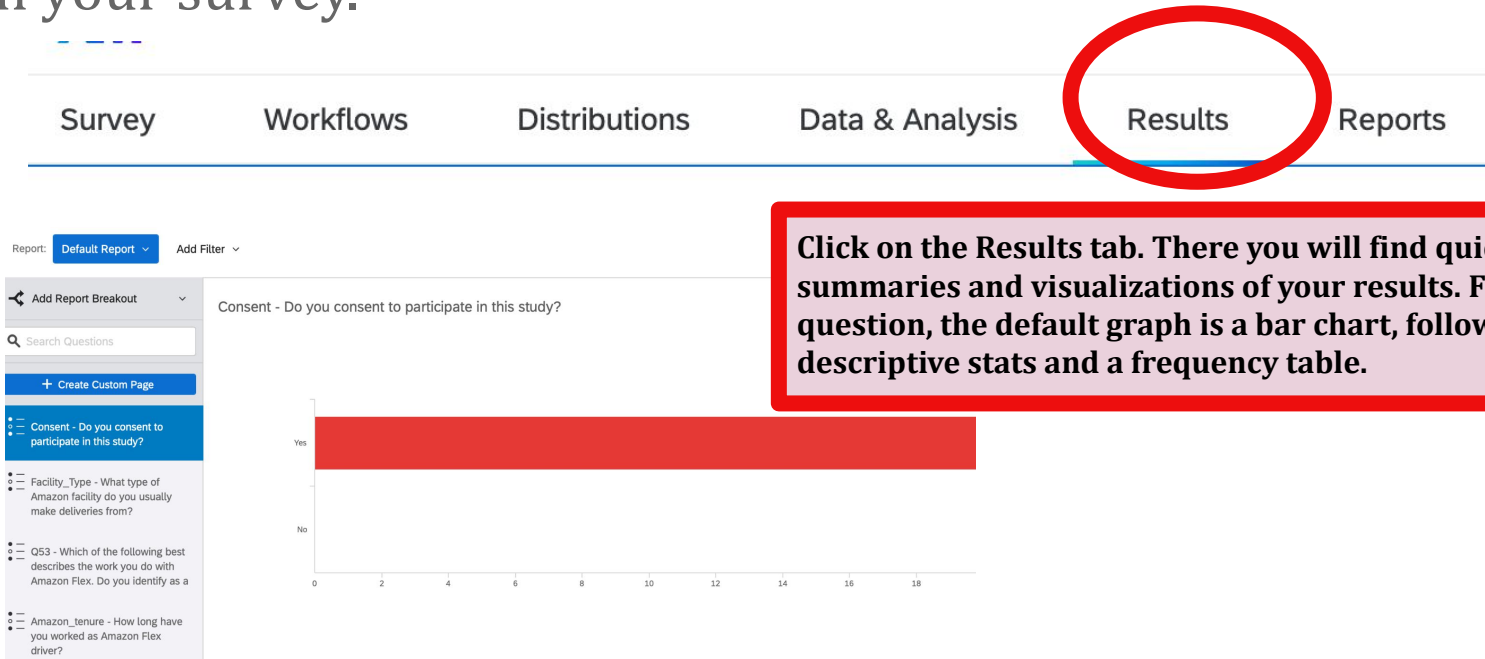
Get a single reusable link



Generate a trackable link for each contact

Analyzing the Results

- The Results tab shows you the basic statistics for each question in your survey.



Click on the Results tab. There you will find quick summaries and visualizations of your results. For each question, the default graph is a bar chart, followed by descriptive stats and a frequency table.

Analyzing the Results

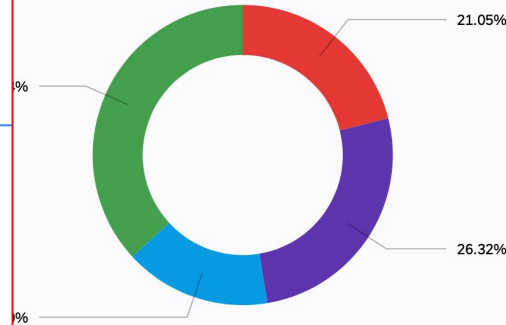
#	Field
1	Less than 10
4	11 to 20 hours
5	21 to 30 hours
6	31 to 40 hours
7	41 to 50 hours
8	50 hours or more

Showing rows 1 to 7 of 7

[+ Add Note](#) [+ Add Visualization](#)

Click “Add Visualization”

Additional visualization options will appear on the left side. You can change the chart type, color codes, and many other features.



Visualization

Pie Chart

Inner Radius

Grouping Percentage Threshold

☒ Show As Percentages

Color Palette

☐ Reverse Colors

Legend Values

☒ Less than 10

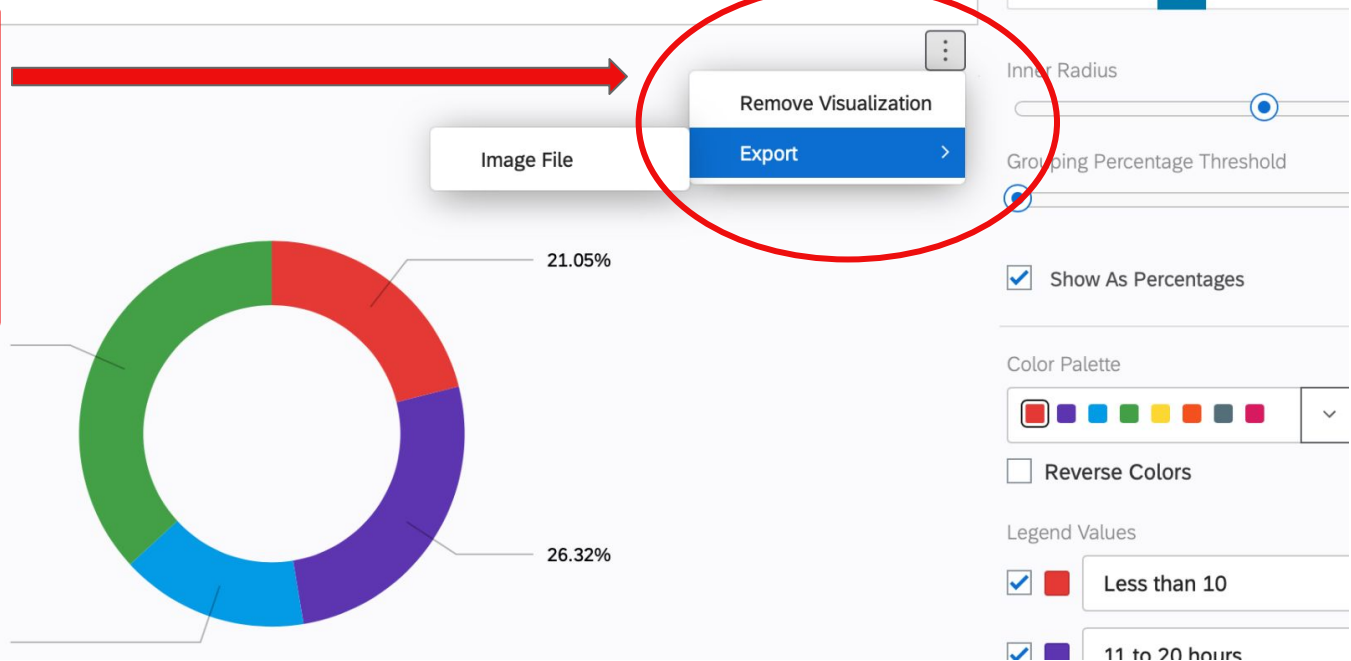
☒ 11 to 20 hours

☒ 21 to 30 hours

Analyzing and Exporting the Results

You can export your chart as an image file. Click on the three vertical dots in the upper left corner of your graph

Showing rows 1 - 7 of 7



Downloading Your Results

Workflows

Distributions

Data & Analysis

Results

Reports

Crosstabs Weighting

Last Record Collected: Never Inactive

Download a data table

CSV TSV Excel XML SPSS Google Drive User-submitted files



Excel

Export your data as an XLSX file - an Excel-compatible format. If you have a very large number of responses, use TSV instead.

[Learn more](#)

☒ Download all fields

☒ Use numeric values

☐ Use choice text

[More options](#)

Close

Download

Export & Import

Export Data...

Import Data...

Response Export Automation...

In “Data & Analysis”, click “Export & Import”. In the window that will pop up, choose the file type you want to work with. If you recoded the answers to numbers and want to download the survey with only numeric values, mark the appropriate field .

Feel free to ask questions at any point during the presentation!

Your Turn!

Activity: Create Surveys

Start creating your surveys! Work individually, in pairs, or in a group. Think about what your overall goal is, what types of experiences you're interested in measuring and tracing, and how your questions can get at those experiences.

1. (if working in a pair or group): **One** person create the Qualtrics survey and share it with everyone. Then, you can all collaborate on that form
2. Include at **least** 10 questions of all different types

Discussion Questions

- What did you find difficult about designing a survey?
- What did you find worked well?
- What do you think is important to keep in mind when you're designing a survey?

Another option: LimeSurvey

LimeSurvey

- Can handle surveys for academic and professional purposes
- Gives access to 800+ pre-designed survey templates
- Allows full control by enabling on-premise deployment, meaning no third-party access to collected data



Why choose LimeSurvey?

Pros

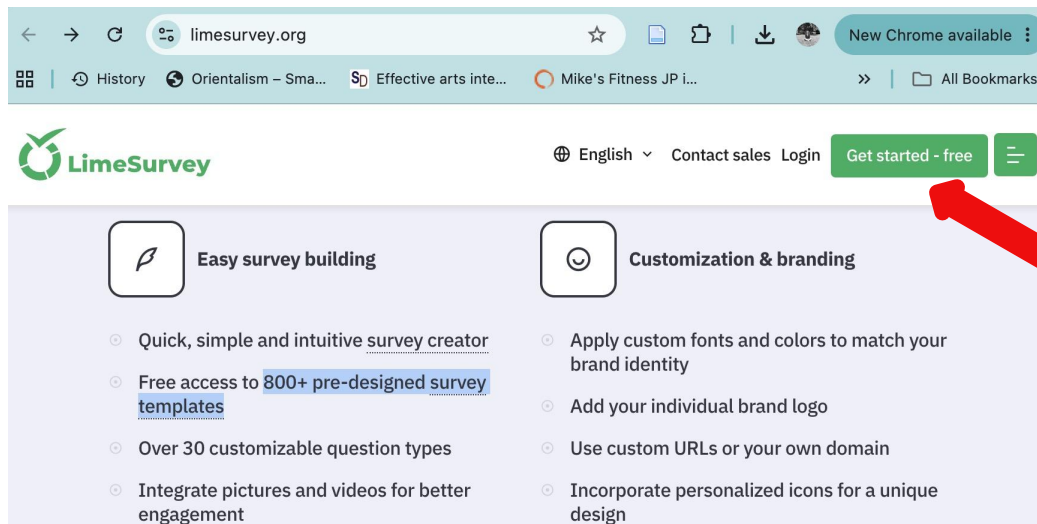
- Free & open source, highly affordable
- Self-hostable, customizable
- No software requirements
- Most mature open-source Qualtrics alternative

Cons

- Mixed support quality
- Weaker analytics/reporting depth
- Basic reports

Getting started

- Go to <https://www.limesurvey.org/>
- Click “Get Started - free”



Sign up

Fill out the information
required to create a free
account



Sign up for free

Already have an account? [Log in](#)

Username

Enter your desired username - min. 6 characters, no spaces.

Password



Enter your desired password - min. 8 characters, 1 uppercase, 1 number.

Email Address

Enter your email address.

Your role



Select a role.

☐

By clicking on "Create account" below, you are agreeing to our [Terms & Conditions](#) and to our [Privacy policy](#).

*Feel free to ask questions at any point
during the presentation!*

Create your survey

Once your account is set up, you can will create a survey site

Survey site is a specific web address and backend environment where your account, database, backups, and questionnaires live

Your survey site

Survey site URL:

Oops! It looks like you have not set up your survey site yet.
Get started today to create and manage your online surveys.

Create survey site



English

S SayyaraP FREE



Survey site URL:

diti.limesurvey.net/admin

View survey site

Domain alias:

Create your survey

Once your survey site is setup click on it

You will see the Welcome to LimeSurvey

Click “Create a new survey”

Welcome to LimeSurvey!



Some piece-of-cake steps to create your very own first survey:

1. Create a new survey by clicking on the  icon.
2. Create a new question group inside your survey.
3. Create one or more questions inside the new question group.
4. Done. Test your survey using the  icon.

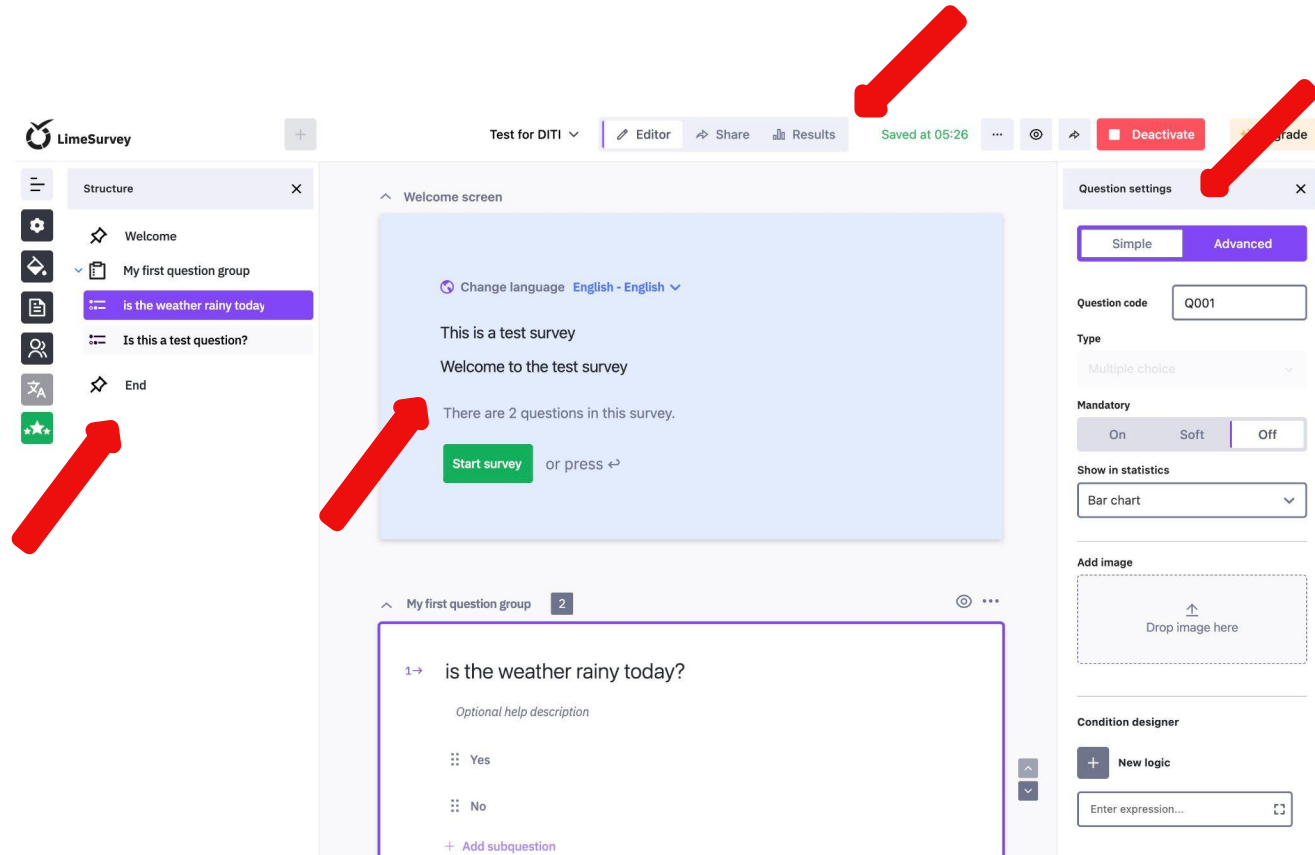
Close

Create a new survey



Vocabulary

1. Canvas
2. Structure panel
3. Setting panel
4. Survey toolbar



Get started

- Name your survey
- Describe your survey
- Start creating questions

Untitled survey



Editor



Share



Results

Saved at 05:12

This is a test survey
Welcome to the test survey

There is 1 question in this survey.

[Start survey](#) or press ↵

My first question group 1

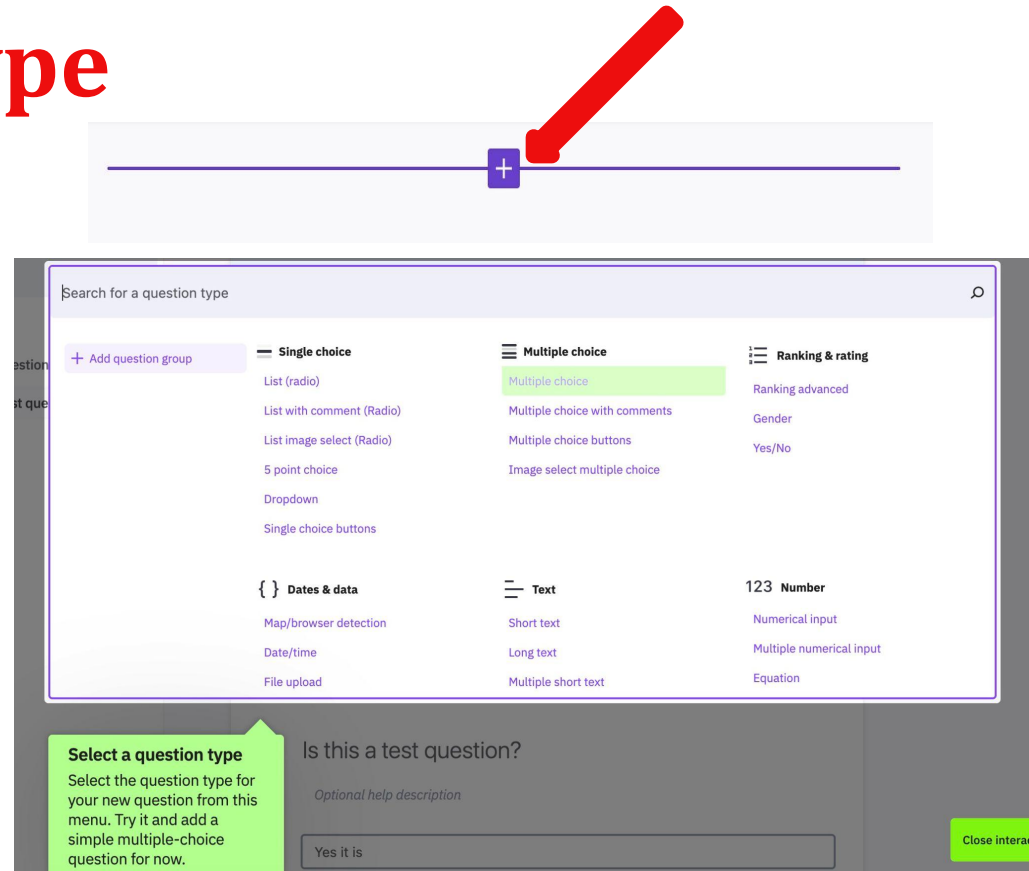
1→ Your question here

Optional help description

+ Add subquestion

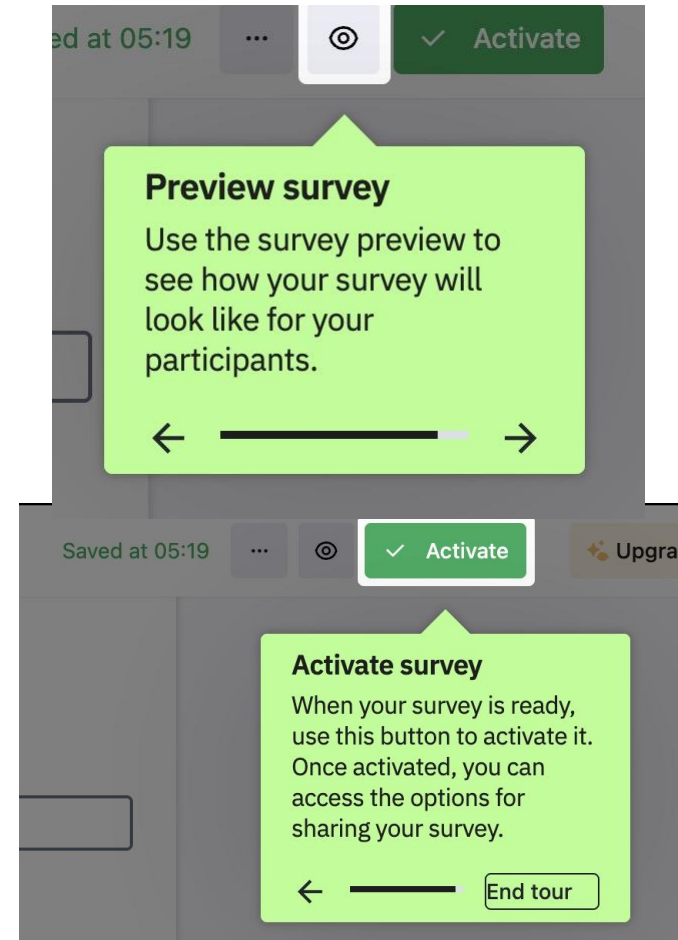
Select question type

- Select “Add contents” button (+ sign)
- Choose an appropriate question type
- Adjust Question Settings if needed



Activate your survey

- Preview your survey
- Select the “Activate” button to activate your survey



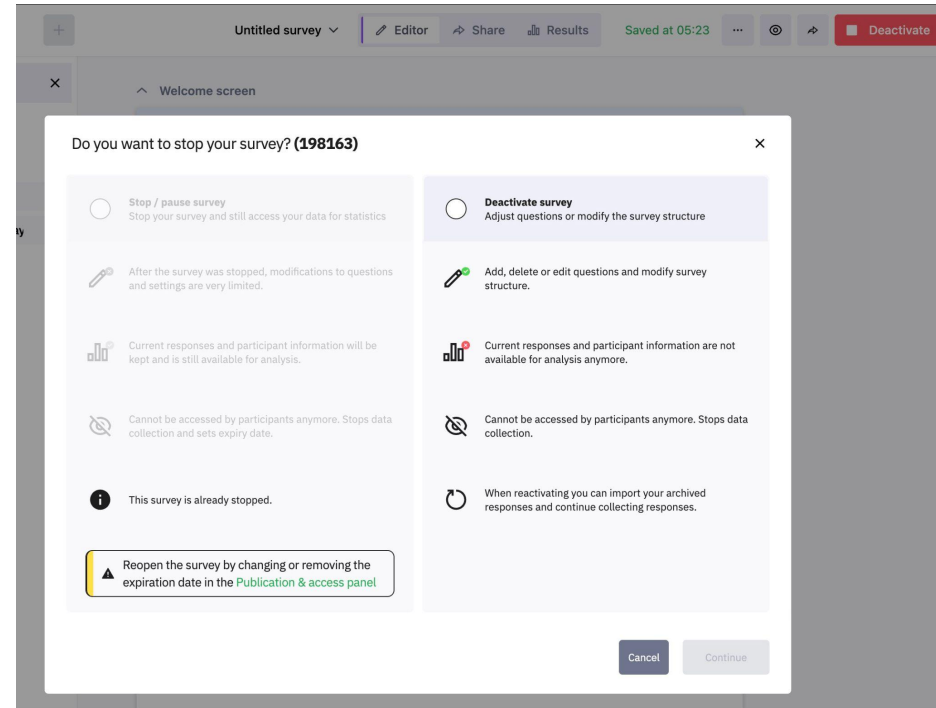
Share your survey

Use the shareable link or the auto generated QR code to share your survey with respondents

The screenshot displays a survey dashboard with a light purple background. At the top, a green notification bar states: "🎉 Congratulations! Your survey has been activated." To the right, the survey is titled "Untitled survey" with a date of "8/14/2026" and a close button (X). Below the notification, three white boxes show response statistics: "0 Total responses", "0 Full responses", and "No responses". A larger white box below these shows "0 Incomplete responses". A purple link "View results overview →" is positioned below the statistics. The main section is titled "Select how you want to share your survey". It features a globe icon and the text "Anyone with link" with a dropdown arrow. Below this, it says "Anyone with the link to this survey can access." A text input field contains a partial URL "diti.limesurvey.ne...?lang=en&newtest=Y", and a purple "Copy" button is next to it. To the right of the text field is a large QR code with a green checkmark in the center. Below the QR code is a "Download QR code" link with a download icon. At the bottom of the main section is a purple link "View sharing overview →". On the right side of the dashboard, there are three buttons: a purple "Edit survey" button with a pencil icon, a red "Deactivate" button with a square icon, and a grey "Preview survey" button with a magnifying glass icon.

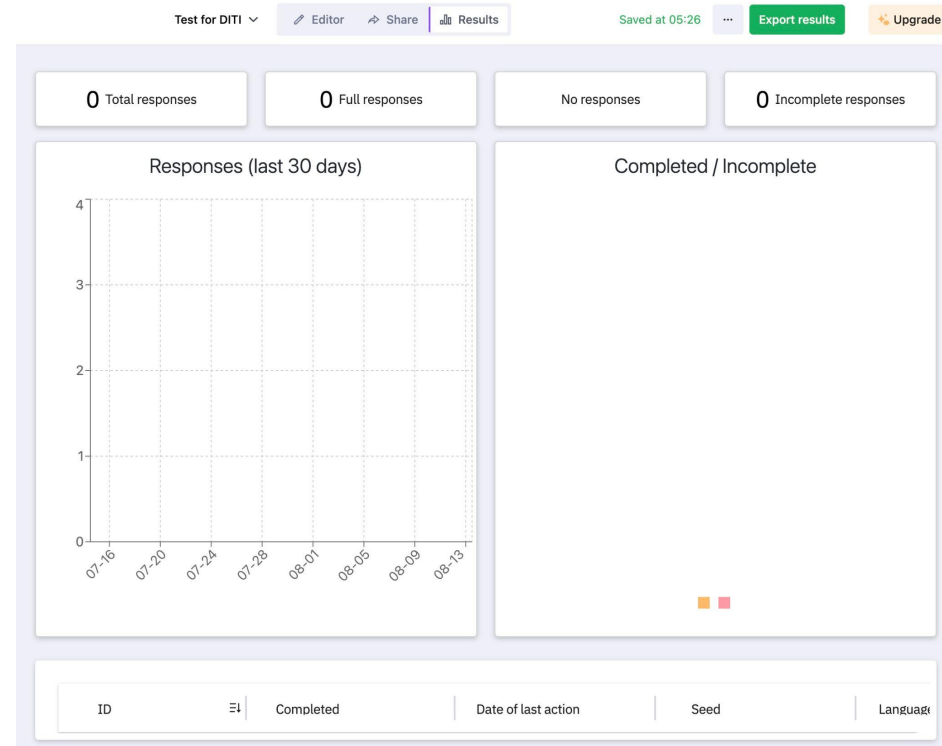
Making changes

- Deactivate your survey to enable the Edit menu
- Hit “Continue” in the pop-up menu



Evaluate the results

In the Survey Toolbar select
“Results” button



Thank you!

Developed by DITI Research and Teaching Fellows: Yana Mommadova, Vaishali Kushwaha, Emily Sullivan, Hunter Moskowitz, Sayyara Poliferno, Mel Williams

- Please fill out this survey to give us feedback:
<https://bit.ly/diti-feedback>
- For more information on DITI, please see: <https://bit.ly/diti-about>
- Schedule an appointment with us! <https://bit.ly/diti-meeting>
- If you have any questions, contact us at: nulab.info@gmail.com